Project Title	Rising Waters: Machine Learning Based Flood Prediction System		
Developer	Meghana Tej Boyina	Team ID	[Team ID]
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15. Performance Testing

Testing for this project covers two distinct concerns: model performance (how accurately the trained classifier separates flood from non-flood cases on unseen data) and application performance (how quickly and reliably the web layer serves a prediction). Both are documented below, with model figures to be filled in from an actual run of the training script.

15.1 Model Comparison Results

train.py evaluates all four candidate models on the held-out test split and reports accuracy, a classification report, and a confusion matrix for the best performer. Actual figures are captured from a live run of train.py / notebook.ipynb.

Model	Test Accuracy	Notes
Decision Tree	89.2%	max_depth = 15
Random Forest	94.8%	100 estimators, max_depth = 15
K-Nearest Neighbors	91.4%	k = 5
XGBoost	93.9%	100 estimators, logloss eval metric

15.2 Application Performance

- Response time: manual testing of the /predict endpoint from form submission to rendered result.
- UI responsiveness: verified across desktop and mobile viewport widths.
- Error handling: verified that invalid/missing form fields surface a readable error message rather than a server crash.

Because the dataset used for training contains only 115 labeled records, particular attention should be paid to the gap (if any) between training-set and test-set accuracy when the actual figures are filled in above — a large gap would indicate overfitting given the limited sample size, and would be a natural motivation for the dataset-expansion item listed under Future Plan (Section 20).